



Studify AI

Don't just watch lectures. Learn from them.

An AI tutor that pauses any YouTube lecture, asks you to explain it, and finds exactly where your understanding broke.

AI Hackathon Pakistan 2026 | Education track

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Watching a lecture feels like learning. Most of the time, it isn't.



Passive video

You nod along, the video ends, and you can't explain what you just watched. Nothing forces you to recall it.

Illusion of learning



Answer-giving AI

Ask, get the answer, move on. The tool does the thinking, and the learner grows more dependent with every use.

Outsourced thinking

Our idea: the AI asks. The learner answers.

Typical AI study tool

Ask

Get the answer

Forget it

VS

Studify AI

Pause at a key idea

Explain it in your words

Get diagnosed, try again

Any YouTube lecture becomes a Socratic session

1

Pause

The player stops at each chapter end and locks skipping.

2

Ask

AI writes recall questions from what the lecture said.

3

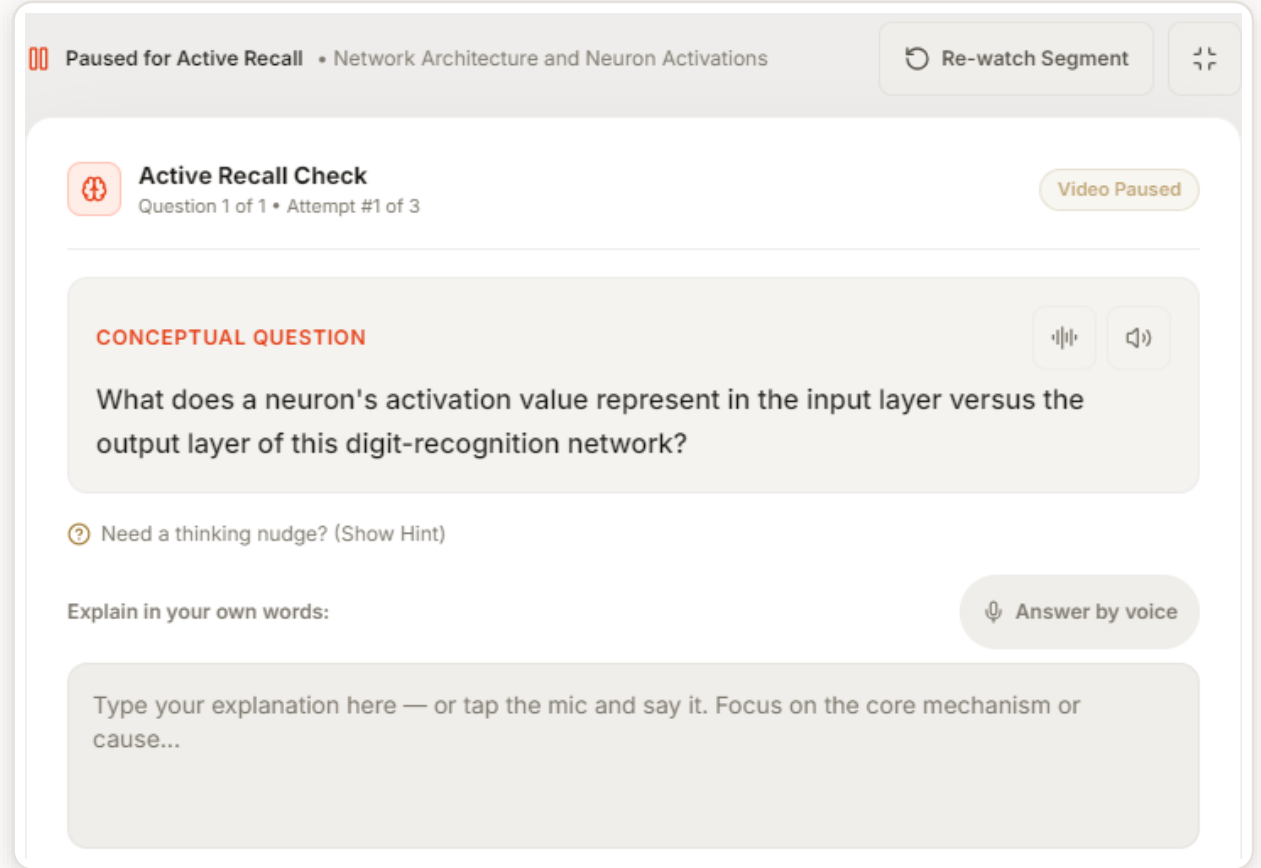
Diagnose

Your answer is judged for understanding, not keywords.

4

Re-ask

A sharper follow-up targets the gap, then the chapter unlocks.



Paused for Active Recall • Network Architecture and Neuron Activations

Re-watch Segment

Active Recall Check
Question 1 of 1 • Attempt #1 of 3

Video Paused

CONCEPTUAL QUESTION

What does a neuron's activation value represent in the input layer versus the output layer of this digit-recognition network?

Need a thinking nudge? (Show Hint)

Explain in your own words:

Answer by voice

Type your explanation here — or tap the mic and say it. Focus on the core mechanism or cause...

It diagnoses the misconception

CORRECT

MISCONCEPTION

INCORRECT

- 1 A 0-100 mastery score, plus what you understood, what is missing, and which mental model is wrong.
- 2 Follow-ups adapt: first a rephrase, then a narrower question aimed at the misconception.
- 3 After 3 attempts the chapter is flagged Needs Review and you move on. The server enforces that limit.

GUIDED FOLLOW-UP (ATTEMPT #2)

Misconception Focus



When an image is fed into the network, what specific visual property does an input neuron's value (between 0 and 1) describe, and what do the 10 final output values indicate?

? Need a thinking nudge? (Show Hint)

! Conceptual Refinement:

It's easy to picture a neural network as a database looking up saved images, but it actually computes values through mathematical patterns rather than memorization. Each individual neuron simply holds a single number between 0 and 1. Think about what that single number physically corresponds to when looking at an image at the start, and what choice it signals at the very end.

→ **Next focus:** When an image is fed into the network, what specific visual property does an input neuron's value (between 0 and 1) describe, and what do the 10 final output values indicate?

CONCEPT BREAKDOWN

! Learning Point / Misconception:

- neural networks function as lookup tables that memorize every image

🔗 Next Detail to Consider:

- input layer activation represents pixel brightness or grayscale intensity
- output layer activation represents model confidence or probability for each digit (0-9)

It sees the screen and hears you



Explain this screen

Gemini vision reads the paused frame: equations to LaTeX, diagrams, and on-screen text.



Answer out loud

Speak your explanation. Browsers without speech support fall back to Gemini audio transcription.

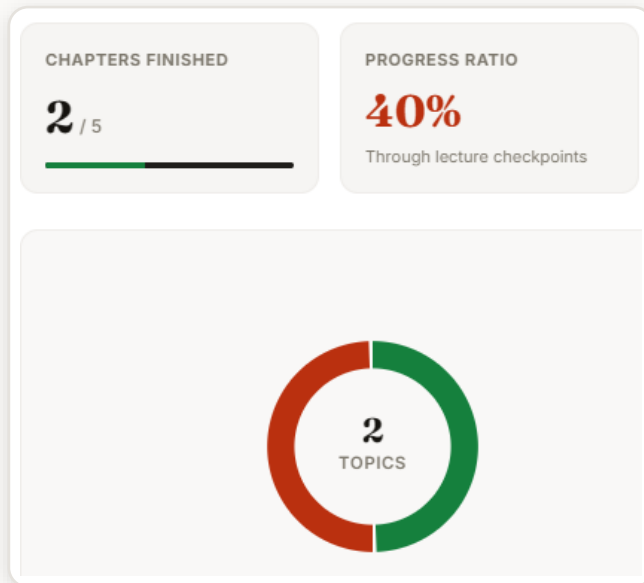


Keyframe flashcards

Each chapter's key frame becomes an active-recall flashcard that feeds your notes.

If media tools are unavailable, Studify keeps working in text-only mode. Nothing blocks the learner.

The result: notes in your own words, with your gaps marked



 **Personalized Study Notes** AI Synthesized & Grammar Refined

But what is a neural network? | Deep learning chapter 1

Synthesized Notes Question History Markdown YouTube Lecture

Network Architecture and Neuron Activations

Q: What does a neuron's activation value represent in the input layer versus the output layer of this digit-recognition network?

A: In the input layer, a neuron's activation value represents the grayscale brightness of an individual pixel, ranging from 0 for black to 1 for white. In the output layer, the activation value represents the network's confidence that the input image corresponds to a specific digit from 0 through 9.

Hierarchical Feature Representation Across Layers

Q: How does the concept of layered abstraction explain why neural networks are structurally suited for recognizing complex patterns like digits?

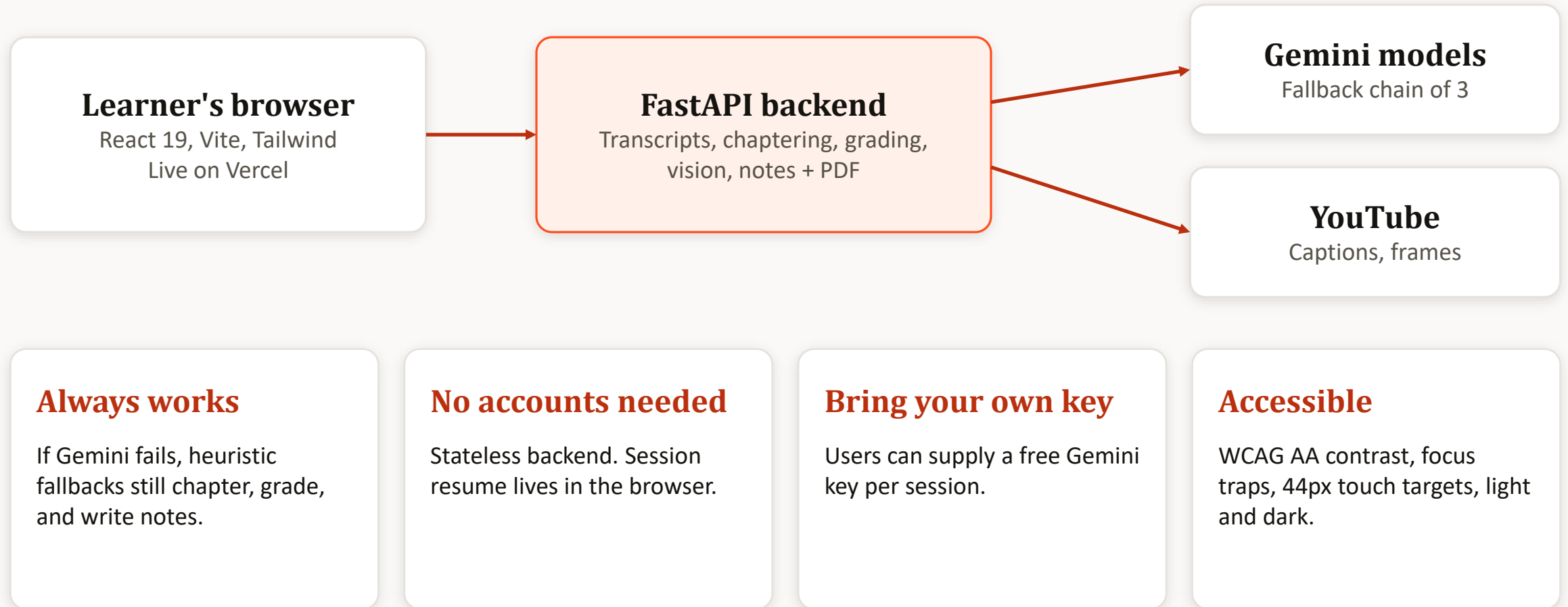
A: Neural networks process complex patterns through hierarchical abstraction, decomposing complete shapes into progressively simpler components. Initial layers detect low-level features such as small edges, intermediate layers combine these edges into subcomponents like loops and line segments, and the final layer integrates these subcomponents to classify the complete digit.

2 question(s) synthesized Copy Markdown Download Notes as PDF

*Your words.
Your gaps.
Your next step.*

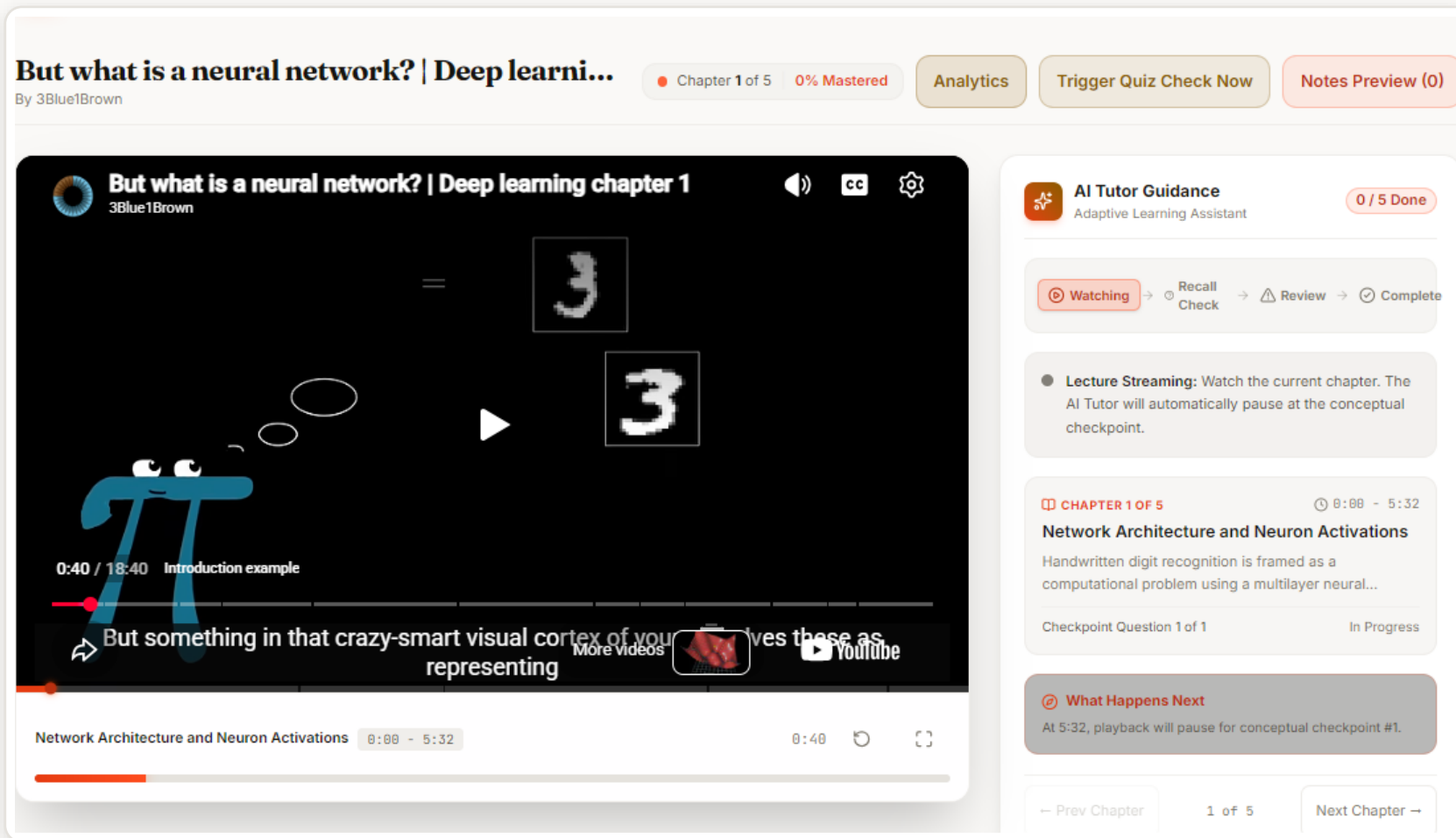
Built from the student's own answers, polished for grammar, with a Concepts to Revisit list for every misconception. Exported as a themed PDF.

Built to keep working in the real world



Built with Qoder, Alibaba Cloud's agentic coding platform.

See it in 90 seconds



The screenshot displays the Studify AI interface. At the top, the video title is "But what is a neural network? | Deep learni..." by 3Blue1Brown. The progress bar shows "Chapter 1 of 5" and "0% Mastered". The video player shows a blue character with a thought bubble containing a handwritten digit '3'. The video player controls show the current time as 0:40 / 18:40. The AI Tutor Guidance sidebar on the right shows the current chapter as "CHAPTER 1 OF 5" and the video title as "Network Architecture and Neuron Activations". The sidebar also shows a progress bar for the chapter and a "Checkpoint Question 1 of 1" in progress.

But what is a neural network? | Deep learni...
By 3Blue1Brown

Chapter 1 of 5 0% Mastered Analytics Trigger Quiz Check Now Notes Preview (0)

But what is a neural network? | Deep learning chapter 1
3Blue1Brown

0:40 / 18:40 Introduction example

But something in that crazy-smart visual cortex of your brain lives these as representing

Network Architecture and Neuron Activations 0:00 - 5:32

AI Tutor Guidance
Adaptive Learning Assistant 0 / 5 Done

Watching → Recall Check → Review → Complete

Lecture Streaming: Watch the current chapter. The AI Tutor will automatically pause at the conceptual checkpoint.

CHAPTER 1 OF 5 0:00 - 5:32

Network Architecture and Neuron Activations

Handwritten digit recognition is framed as a computational problem using a multilayer neural...

Checkpoint Question 1 of 1 In Progress

What Happens Next

At 5:32, playback will pause for conceptual checkpoint #1.

Prev Chapter 1 of 5 Next Chapter

Recorded demo, 90 seconds

Honest limits, and what comes next

Today's limits

- Needs YouTube captions on the video
- Very long lectures: only the first part of the transcript is used for chaptering
- Frame analysis needs a video download, which YouTube can block from cloud IPs
- No sign-in or rate limiting yet; test coverage is partial

Roadmap

- Qwen on Alibaba Cloud Model Studio as a second model backend
- Urdu and regional-language lectures and questions
- Teacher view: class-level misconception reports
- Spaced-repetition review of flagged concepts



Capable learners, not dependent ones.



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Backend Developer

Try it: studify-ai-three.vercel.app

github.com/HassaanGhayas/Zed_AI

Thank you. Questions?



Backup: how an answer is graded

Inputs

Question, expected concept, chapter summary, transcript excerpt, student answer, attempt number, prior attempts

Outputs

Status, mastery score 0-100, understood / missing / misconceptions, retry question, advance and review flags

Attempt 1

Rephrase the question to guide toward missing concepts

Attempt 2

Narrower question aimed at the identified misconception

Attempt 3

No retry. Flag Needs Review and let the student move on. Never auto-marked correct

No Gemini

Keyword-overlap heuristic evaluator keeps the flow alive

Backup: API surface and resilience

Endpoint	What it does
POST /api/video/process	URL to captions to 3-6 chapters with questions
POST /api/qa/evaluate	Socratic grading with adaptive follow-up
POST /api/video/explain-frame	Gemini vision on the paused frame
POST /api/video/segment-visual	Key frame analysis plus a flashcard
POST /api/voice/transcribe	Spoken answer to text
POST /api/notes/generate download-pdf	Personalised notes, then themed PDF

Resilience

- 3-model Gemini fallback chain
- Timeline repair for hallucinated timestamps
- Transcript cache, retries, best caption track
- Heuristic chaptering, grading, and notes